1. Feifei's quorum sensing model

Particles diffuse faster when their perceived crowding exceeds a threshold.

$$\mathrm{d}\mathbf{r}_i = \sqrt{2D(P_i)}\,\mathrm{d}\mathbf{W}_i\,, \qquad P_i = \sum_{\substack{j\neq i \\ 0<r_{ij}<d_c}} \frac{1}{2\pi r_{ij}}\,.$$

$$D(P) = \begin{cases} D_{\min}, & P \le P_{\text{th}}, \\ D_{\max}, & P > P_{\text{th}}, \end{cases} \qquad 0 < D_{\min} < D_{\max}.$$

Nonlocal perception: an unnormalized, isotropic sensing kernel

$$K(r) = \frac{\Theta(d_c - r)}{2\pi r}, \qquad P = K * \rho, \qquad \int K(r)\,\mathrm{d}^2 r = d_c.$$

Itô expansion and random-current covariance give the density equation

$$\partial_t \rho = \nabla^2[\rho D(P)] - \nabla\cdot\left[\sqrt{2\rho D(P)}\,\Lambda\right].$$

Λ is unit vector white noise. Self-sensing is excluded.

2. Linear fluctuations and the structure factor

Assume a smooth response near the uniform state

$$\rho = \rho_0 + \delta\rho, \qquad \overline{P} = \rho_0 d_c, \qquad \overline{D} = D(\overline{P}) > 0, \qquad g = D'(\overline{P}).$$

$$\partial_t \delta\rho = \nabla^2 \left[\overline{D}\, \delta\rho + \rho_0 g (K * \delta\rho) \right] - \nabla \cdot \left[\sqrt{2\rho_0 \overline{D}}\, \Lambda \right].$$

Each nonzero Fourier mode is a scalar Ornstein–Uhlenbeck process

$$\partial_t \delta\rho_{\mathbf{k}} = -\lambda(k) \delta\rho_{\mathbf{k}} + \eta_{\mathbf{k}}, \qquad \lambda(k) = k^2 \left[\overline{D} + \rho_0 g \widehat{K}(k) \right].$$

$$\underbrace{Q(k) = 2\rho_0 \overline{D} k^2}_{\text{noise power}}, \qquad \underbrace{2\lambda(k) C(k) = Q(k)}_{\text{stationary variance}}, \qquad C(k) = \langle |\delta\rho_{\mathbf{k}}|^2 \rangle.$$

$$S(k) = \frac{C(k)}{\rho_0} = \frac{\overline{D}}{\overline{D} + \rho_0 g \widehat{K}(k)}$$

Valid for stable modes, $\overline{D} + \rho_0 g \widehat{K}(k) > 0$, within the linear-noise approximation.

3. Why an approximately linear spectrum appears

The 2D Fourier transform reduces to a Bessel integral

$$\widehat{K}(k) = \int_0^{d_c} J_0(kr)\,\mathrm{d}r = \frac{F(kd_c)}{k}, \qquad F(x) = \int_0^x J_0(u)\,\mathrm{d}u.$$

Define $k_* = \rho_0 g/\overline{D} > 0$ and $B = k_* d_c$. Then

$$S(k) = \left[1 + \frac{k_*}{k} F(kd_c)\right]^{-1}.$$

For $kd_c \gg 1$, $F(kd_c) \simeq 1$ and $\widehat{K}(k) \simeq 1/k$.

$$\underbrace{B \gg 1}_{\text{For } P \to P_{\text{th}},\ D(P) \text{ has a sharp transition } g \to \infty,\ B \to \infty}, \quad d_c^{-1} \ll k \ll k_* \quad \Longrightarrow \quad S(k) \simeq \frac{k}{k_*}$$

Mechanism: the sensing kernel changes the relaxation scaling

$$\lambda(k) \simeq \rho_0 g k, \qquad Q(k) \propto k^2, \qquad S(k) = \frac{Q(k)}{2\rho_0 \lambda(k)} \propto k.$$

This intermediate range must lie below the microscopic and continuum cutoffs.

4. Finite-range limit and the threshold approximation

At fixed finite sensing radius, the true small-k limit is a plateau

$$\widehat{K}(k) = d_c\left[1 - \frac{(kd_c)^2}{12} + \cdots\right], \qquad \lim_{k\to 0} S(k) = \frac{1}{1+B} > 0.$$

A finite-range $S(k) \sim k$ window alone does not establish asymptotic hyperuniformity.

The microscopic step needs an explicit smooth or effective response

$$D_\varepsilon(P) = D_{\min} + \frac{D_{\max} - D_{\min}}{1 + \exp[-(P - P_{\text{th}})/\varepsilon]}.$$

$$g = D'_\varepsilon(\overline{P}) > 0 \text{ near the switching region}, \qquad |\delta P| \ll \varepsilon.$$

For the literal step, $g = 0$ away from threshold and is undefined at threshold.

Thus the prediction is a smooth-response or effective-closure result.